He ZiAng

System Engineer | Backend APIs & System Integration

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Professional summary

System Engineer developing manufacturing APIs and integration services since 2022 using Node.js/Express, Python, Oracle, and PostgreSQL. Experience includes event-driven equipment monitoring, database-backed workflows, application troubleshooting, Docker-based UAT preparation, and React interfaces.

Work experience

Resonac HD Singapore 2022.03 - Present
System Engineer Singapore

- Developed Node.js/Express APIs backed by Oracle stored procedures for a measurement-data platform, supporting engineering configuration, data processing, and result review. Built Python ingestion and Excel reporting workflows and React interfaces.
- Built a Bun/Express visual inspection API with route permissions, transactional Oracle writes, audit history, and inspection and reporting workflows, alongside a React/TypeScript frontend. Prepared Docker-based UAT deployment with Oracle and inspection-image integration.
- Developed Python equipment-log pipelines configured for multiple production lines, converting manufacturing logs into Parquet datasets, extracting process cycles, tracking Oracle metadata, and archiving data. Added database connection pooling, processing diagnostics, and Flask visualization applications.
- Developed and supported equipment-monitoring services using Python, PostgreSQL, MQTT, and web APIs for pump condition, motor torque, and process parameters. Implemented event-driven threshold alerts, PLC-connected cassette tracking, WebSocket broadcasts, and exception-log ingestion.
- Extended an ASP.NET/VB.NET measurement-management system with instrument configuration, requests, acknowledgements, result uploads, and statistical process control (SPC) integration. Added instrument workflows and contributed to measurement reporting.
- Contributed to migrating manufacturing alert integrations from Workplace to Microsoft Teams. Updated application integrations, caller configuration, and message formatting through a shared Microsoft Graph service.
- Used LLM coding assistants, including Codex and Claude Code, for development, refactoring, debugging, and test preparation, with review and verification of generated changes.

Vanitee 2017.09 - 2018.02
iOS Developer Intern Singapore

- Worked on a Swift business application for iPhone and iPad and integrated React Native into the iOS application to support code sharing with Android.

Technical skills

Backend and APIs: Node.js, Express, Flask, Bun, ASP.NET, REST APIs, API authorization.

Databases and integration: Oracle, PostgreSQL, stored procedures, connection pooling, MQTT, WebSockets, SPC integration.

Languages and data processing: Python, JavaScript, TypeScript, SQL, VB.NET, pandas, Parquet; earlier Swift and Java experience.

Delivery and interfaces: Git, Docker, UAT preparation, application logging, troubleshooting, React.

Education

University of New South Wales 2018 - 2020
Computer Science Bachelor's degree

Ngee Ann Polytechnic 2015 - 2018
Information Technology Diploma

Languages

Chinese [Native] / English [Professional]